

Matgeo Presentation - Problem 4.11.31

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Question

Find the area of the region bounded by line $y=3x+2$, the X axis and the ordinates $x=-2$ and $x=1$.

Solution

let

$$\mathbf{A} = \begin{pmatrix} -2 \\ 0 \end{pmatrix} \quad (0.1)$$

$$\mathbf{C} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (0.2)$$

$$(0.3)$$

let $\mathbf{D}$ and $\mathbf{E}$ be the vectors on the line corresponding to $x = -2$ and $x = 1$
Given line equation is

$$-3x + y = 2 \quad (0.4)$$

which can be expressed as

$$\mathbf{n}^T \mathbf{x} = c \quad (0.5)$$

$$\Rightarrow \begin{pmatrix} -3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \quad (0.6)$$

$$\mathbf{n} = \begin{pmatrix} -3 \\ 1 \end{pmatrix} \text{ and } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \text{ and } c = 2 \quad (0.7)$$

Solution

let us find the vector **D**

$$(-3 \quad 1) \begin{pmatrix} -2 \\ y \end{pmatrix} = 2 \quad \Rightarrow y = -4 \quad (0.8)$$

$$\Rightarrow \mathbf{D} = \begin{pmatrix} -2 \\ -4 \end{pmatrix} \quad (0.9)$$

as $y < 0$ we should find the **B** where the line meets the x axis

$$(-3 \quad 1) \begin{pmatrix} x \\ 0 \end{pmatrix} = 2 \quad \Rightarrow 3x = -2 \quad (0.10)$$

$$\Rightarrow \mathbf{B} = \begin{pmatrix} -\frac{2}{3} \\ 0 \end{pmatrix} \quad (0.11)$$

let us find the vector **E**

$$(-3 \quad 1) \begin{pmatrix} 1 \\ y \end{pmatrix} = 2 \quad \Rightarrow y = 5 \quad (0.12)$$

$$\Rightarrow \mathbf{E} = \begin{pmatrix} 1 \\ 5 \end{pmatrix} \quad (0.13)$$

Conclusion

The area to be computed is area of $\triangle EBC$ + area of $\triangle ABD$

$$ar(\triangle ABD) = \frac{1}{2} \|(A - B) \times (A - D)\| \quad (0.14)$$

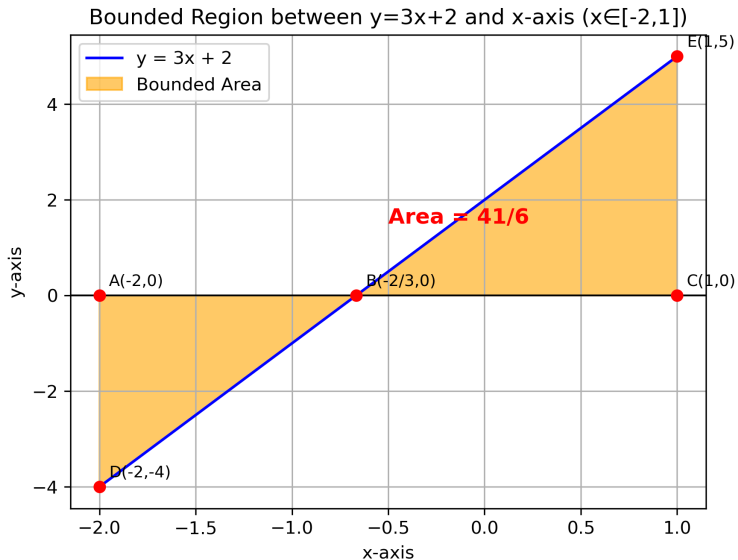
$$= \frac{1}{2} \left\| \begin{pmatrix} 4 \\ 3 \end{pmatrix} \times \begin{pmatrix} 0 \\ 4 \end{pmatrix} \right\| = \frac{8}{3} \quad (0.15)$$

$$ar(\triangle EBC) = \frac{1}{2} \|(E - c) \times (B - c)\| \quad (0.16)$$

$$= \frac{1}{2} \left\| \begin{pmatrix} 0 \\ 5 \end{pmatrix} \times \begin{pmatrix} -5/3 \\ 0 \end{pmatrix} \right\| = \frac{25}{6} \quad (0.17)$$

$$\Rightarrow \text{area of the region is } = \frac{8}{3} + \frac{25}{6} = \frac{41}{6} \quad (0.18)$$

Plot



C Code: area.c

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h> // for fabs()

int main() {
    FILE *fp;
    double area;

    // Compute area = from -2 to 1 |3x+2| dx
    // Split into two parts: [-2, -2/3] and [-2/3, 1]

    double x0 = -2.0/3.0;

    // Antiderivative F(x) = (3/2)x^2 + 2x
    double F_at_x0 = (1.5 * x0 * x0) + (2.0 * x0);
    double F_at_neg2 = (1.5 * (-2) * (-2)) + (2.0 * (-2));
    double F_at_1 = (1.5 * 1 * 1) + (2.0 * 1);

    // Left interval [-2, -2/3] -> integrand is negative
    double left = -(F_at_x0 - F_at_neg2);

    // Right interval [-2/3, 1] -> integrand is positive
    double right = (F_at_1 - F_at_x0);

    area = left + right; // total geometric area

    // Open file area.dat
    fp = fopen("area.dat", "w");
    if (fp == NULL) {
        printf("Error opening file!\n");
        return 1;
    }
}
```

C Code: area.c

```
// Write output
fprintf(fp, "Line equation in matrix form:  $[3 \times 1] [x \ y]^T = -2$ \n");
fprintf(fp, "Area of the region = %.6f (exact = 41/6)\n", area);

fclose(fp);

printf("Area successfully written to area.dat\n");
return 0;
}
```


Python: plot.py

```
import matplotlib.pyplot as plt
import numpy as np

# Define line
x = np.linspace(-2, 1, 400)
y = 3*x + 2

# Vertices
A = (-2, 0)
B = (-2/3, 0) # x-intercept
C = (1, 0)
D = (-2, -4)
E = (1, 5)

# Plot line
plt.plot(x, y, 'b-', label="y=3x+2")

# Plot x-axis
plt.axhline(0, color='black', linewidth=1)

# Shade the bounded area (polygon A-B-C-E-D-A)
polygon_x = [A[0], B[0], C[0], E[0], D[0], A[0]]
polygon_y = [A[1], B[1], C[1], E[1], D[1], A[1]]
plt.fill(polygon_x, polygon_y, color='orange', alpha=0.6, label="BoundedArea")

# Mark and label vertices (B shown symbolically)
vertices = {
    "A(-2,0)": A,
    "B(-2/3,0)": B, # keep symbolic form
    "C(1,0)": C,
    "D(-2,-4)": D,
    "E(1,5)": E
}
```

Python: plot.py

```
for name, (px, py) in vertices.items():
    plt.plot(px, py, 'ro')
    plt.text(px+0.05, py+0.2, name, fontsize=9)

# Annotate area
plt.text(-0.5, 1.5, "Area= $\frac{41}{6}$ ", fontsize=12, color="red", weight="bold")

# Labels and formatting
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Bounded Region between  $y=3x+2$  and x-axis ( $x \in [-2, 1]$ ")
plt.legend()
plt.grid(True)

# Save figure
plt.savefig("bounded_area_vertices.png", dpi=300)
plt.show()
```